

Answers

VARC

1. 4	2. D	3. A	4. A	5. A	6. 2143	7. A	8. B
9. D	10. B	11. C	12. B	13. 3421	14. C	15. A	16. B
17. D	18. A	19. 1	20. D	21. A	22. C	23. A	24. A

LRDI

25. 4	26. D	27. A	28. B	29. 4	30. 0	31. 5	32. 4
33. D	34. C	35. B	36. B	37. D	38. B	39. 50	40. 80
41. 40	42. 60	43. 7	44. D	45. A	46. 6		

Quant

47. B	48. D	49. 6	50. A	51. 15	52. A	53. D	54. 16
55. A	56. D	57. 55	58. A	59. A	60. A	61. 900	62. C
63. D	64. C	65. D	66. 60	67. 3	68. 175		

Explanations

VARC

1. 4

All five sentences revolve around a philosophical discussion of freedom over death: what it means, why it matters, and how it shapes debates like assisted dying. Most of the sentences develop an argument about freedom as a moral and philosophical idea, not just popular opinion.

Sentences 5, 1, 2, and 3 follow a clear progression. Sentence 5 opens the discussion by noting that freedom is philosophically complex and sets a thoughtful tone. Sentence 1 then focuses on a specific kind of freedom, freedom over death, and places it in a historical and moral context. Sentence 2 builds on this by arguing that this freedom should be central to the broader debate. Sentence 3 then presents legalised assisted dying as a policy or ethical step toward realising this freedom.

Sentence 4 interrupts this flow. Rather than adding to the author's conceptual argument, it makes a descriptive claim about what "many people endorse" in practice. This reference to public opinion is not needed or connected to the philosophical progression of the other sentences. As a result, it feels off-topic instead of supporting the main argument. So, sentence 4 is the one that does not fit.

 VIDEO SOLUTION

2. D

In this passage, in the very first paragraph, the author asks two basic questions about the phenomenon of electronic music. In the following paragraphs, the author tries to give an explanation of these questions.

Let's analyse each and every option.

Option A: The goal of the author is not to differentiate between a modern composer and a nineteenth-century composer. So, we can eliminate this option.

Option B: The author does not differentiate electronic music and other forms, but defends electronic music against some questions. So, we can eliminate this option.

Option C: This is a sub-idea, not the author's main goal. We can eliminate this option.

Option D: This option captures the main goal of the author, as essentially the main idea is to defend electronic music against common charges. So, this is the correct answer.

[VIDEO SOLUTION](#)

3. A

The "communication problem" is stated in the first line of paragraph 2. The author basically expands on this in further lines of the same paragraph. He mentions that new language of forms like 'densities', 'indefinite pitch relations' are used instead of traditional concepts of harmony, melody, rhythm, etc. He further states that once these terminologies are understood, the barrier between the listener and the work will be removed.

Let's analyse the options.

Option A: This option captures the author's idea exactly, and hence it is the correct answer.

Option B: "We cannot see electronic music as music" sounds extreme and hence can be eliminated.

Option C: This is again wrong. There is a relation between the discussions mentioned by the author in the first and second paragraphs. So, we can eliminate this option.

Option D: This is not the correct idea. The communication problem is not something that helps us to see electronic music as music. So, we can eliminate this option.

So, the correct answer is option A.

[VIDEO SOLUTION](#)

4. A

This is a single negation question based on Stravinsky's description in the first paragraph of the passage.

Let's analyse the options.

Option B: "respond to and expand upon earlier understandings of music", this idea is conveyed by Stravinsky's description of music as "a form of speculation in terms of sound and time". So, we can eliminate this option.

Option C: This idea is also conveyed in Stravinsky's description, so we can eliminate it.

Option D: This idea is also conveyed through the lines: "characteristic twentieth-century view of the art of musical composition will of course immediately bring the layman closer to an understanding of,..." So, we can eliminate this option.

Option A: Helping us to determine which sounds are musical and which are not, this idea is not conveyed by Stravinsky's description. So, this is the correct answer.

[VIDEO SOLUTION](#)

5. A

The phrase "sui generis" is mentioned in the last line of the third paragraph of the passage.

The author mentions that the magic of the new medium lies in the exploration of "sui generis" languages and forms.

So, from here, we can understand that the author is referring to certain unique or particular languages.

So, the correct answer is option A, particular.

Option-B (generic), option-C (unaesthetic), and option-D (indescribable) do not fit the context of "sui generis" language and hence can be eliminated.

So, option A is the correct answer.

[VIDEO SOLUTION](#)



6. 2143

The four sentences together explain what the panoptic mechanism is and how it works. First, they place it in the political sphere, then show how it connects power to social functions, describe its technical effects, and finally sum up the logic in a clear, conceptual way.

To find the right order, we start with sentence 2. Although informal, it sets the stage by placing the topic in the political sphere and showing that the idea makes sense from this view. Sentence 1 comes next, explaining what the mechanism does and how it fits into areas like education, medicine, punishment, and production. It also shows how power and knowledge are adjusted for supervision and productivity. Sentence 4 then sums up the explanation. It starts with "In short," and explains that power is built in, not forced from outside, to make things more efficient. Sentence 3 is last and gives the most abstract definition, showing that the panoptic mechanism is not just a link between power and function, but a way for each to work through the other. The sentences move from a 'general assessment' to a more 'formal definition.' Hence, the correct order is **2143**.

[VIDEO SOLUTION](#)

7. A

The paragraph explains how the Delgado sisters stay committed to traditional instrument-making. It starts with a general statement about inherited methods, gives specific examples, and then discusses the craft's success and legacy over generations.

The missing sentence is a direct quote that highlights the use of *old-world, traditional techniques from Mexico*. It fits best right after the opening line, which says the sisters “embrace the ways their great-grandfather built and repaired instruments.” Putting the quote at blank (1) lets Ava’s words explain what “embracing those ways” means before the paragraph goes into details about materials, tools, and construction methods.

If the quote is placed at blank (2) or (3), it would break up the flow after the technical details or after talking about commercial success. At blank (4), it would be too late, since the focus has already moved to fame and childhood involvement, making the quote feel out of place. Therefore, the sentence best fits at position 1 (Option 1).

VIDEO SOLUTION

8. B

The paragraph discusses the social role and market for silver. It first explains why silver was important, then why that importance faded, and finally what kept the silver trade going. The sentence in question gives historical context by showing why silver mattered: it was a visible symbol in private, religious, and diplomatic settings. This idea helps connect the decline in domestic use to the reasons the trade survived.

Looking at the paragraph structure, after talking about electric lighting and changes in home life, the text says: “*Domestic and economic changes may have worked against the market, but the London silver trade remained buoyant...*” Right before this point, we need a reminder of *what silver’s traditional role had been*, as this makes the decline more meaningful and highlights why the trade’s survival is surprising.

Putting the sentence in position (3) achieves this goal. It looks back at silver’s long-standing role in display, which helps make the later discussion about collectors, museums, and institutional buyers clear and relevant. The sentence does not work as well at (1), since the decline has not been introduced yet. It also does not fit at (2), because that spot comes after a quote about the visual issue. By (4), the paragraph has already moved on to American museum acquisitions. So, the sentence fits best at position 3 (option 3).

VIDEO SOLUTION

9. D

We evaluate each option individually:

Option A is clearly supported by the passage, as we find direct evidence for it. The passage explicitly cites “a run on the banks” and “a scramble to buy toilet paper” as collective settings “where contagion shapes behaviour,” and explains that such systems can generate extreme events beyond normal expectations. This supports the inference that local choices amplify into system-wide cascades.

Option B is also supported. The passage states that once a first-order tail event occurs, “*the rules change, and the probability of a second-order tail event increases*,” and that such events “*change the perceived costs of our actions and changes the rules that we play by*.” This aligns with the claim that a rare shock alters payoffs and raises the likelihood of subsequent disturbances.

Option C is supported as well. The author explains that in heavy-tailed settings, “*there is a much higher probability of extreme events*,” which are “*more frequent and larger than would be expected under normal distributions*.” This directly supports the inference about extreme outcomes becoming more frequent and complicating risk assessment.

Option D, on the other hand, is not supported by the passage. The passage never attributes the rebounds in financial markets solely to sports bettors, and simply states them as ‘a potential cause’ for ‘some of these dynamics’ [... *some of these dynamics are potentially attributable to former sports bettors... entering the market as speculators...*] Option D, therefore mis-represents and exaggerates the claims in the passage. Hence, we can conclude that only Option D is not supported by the passage.

VIDEO SOLUTION

10. B

The passage explains how complex systems act when faced with uncertainty and contagion. It starts by showing that randomness, or “noise,” can sometimes create order, like in fish schools. Next, it describes how systems shaped by imitation or contagion are open to heavy-tailed events, meaning extreme outcomes happen more often than a normal distribution would predict. The passage then discusses how one extreme event can lead to others, since big shocks change incentives, risk perceptions, and behaviour. This is summed up by the idea of nonstationarity, with examples from pandemics, financial markets during COVID-19, and evolutionary arms races.

Option B best summarises this sequence. It includes the opening idea that ‘noise can create order’, includes the main point about heavy-tailed events in contagion-driven systems, and notes that early shocks change the system’s rules through nonstationarity. It also uses the COVID-era market as an example, not the main point, which matches the passage’s focus.

The other options each have problems. Option A goes against the passage by saying social outcomes usually follow normal distributions and that extreme events are rare, while the passage argues the opposite. Option C overgeneralizes the example of the bettors: the passage only suggests that market dynamics may have been affected by displaced sports bettors, not that these entrants *always* cause inefficiency or guarantee profits for long-term investors. Moreover, option C frames the example, or rather the inference from it, as the primary argument in the passage, which is clearly a supporting argument. Option D goes beyond the scope of the passage by incorrectly associating ‘nonstationarity’ with ‘evolutionary biology’, even though the passage only applies it to markets, pandemics, and technology-driven social systems.

VIDEO SOLUTION

CAT Percentile Predictor



11. C

The passage says that in complex systems, a rare but extreme event can change how the system works, making more extreme events likely. This usually happens because incentives, behaviour, or limits change after the first shock. To support this idea, we need to see that after one extreme event, more extreme events happen more often than usual. To identify which option strengthens this claim the most, we evaluate each option individually.

Option A weakens the claim rather than strengthening it. It explicitly states that super-spreading events are isolated spikes and that later outbreak sizes revert to baseline distributions with no increase in extreme clusters. This directly contradicts the idea that first-order tail events raise the probability of subsequent tail events.

Option B does not relate to the passage’s argument. If river discharge stays normal and stable during storms, it shows there is no link between extreme events. This does not support the idea of cascading tail events in complex systems.

Option C strongly supports the passage's claim. It shows that after a major equity crash (a first-order tail event), extreme price movements cluster densely over subsequent weeks, occurring far more frequently than under normal conditions. This directly illustrates second-order tail events: the initial shock changes market dynamics so that further extreme outcomes become more likely, exactly as described in the passage.

Option D also goes against the claim. It describes how seismic activity returns to normal with no aftershocks, which suggests that events are independent and that one extreme event does not make others more likely.

So, the observation that best supports the passage's claim is option C.

VIDEO SOLUTION

12. **B**

The passage explains that when people make sensible choices individually, these actions can add up and create unexpected group results. For contact-tracing apps to increase risky interactions, people's actions need to affect each other rather than being independent. To check if an assumption is necessary for this argument to hold true, we can use the negation test.

Option A says most users stop using the app quickly, so only highly exposed users remain, which stops any big changes in movement patterns. Even if this happens, the main concern of the passage still stands, because the argument is about how people's behaviours interact, not about participation bias cancelling effects. The claim does not depend on early uninstallation stopping feedback, so this assumption is not necessary.

Option B says people change their movement partly because of infection information and partly because of what others do, so many small actions add up. If this were not true, and people made decisions on their own without reacting to others, then local changes from the app would not build into the group patterns the passage warns about. The whole process described in the passage would not work. So, this assumption is necessary.

Option C says the app alerts are always perfectly accurate and real-time for everyone. The passage does not rely on perfect technology; it relies on how people react. Even if the information is not exact or is delayed, it can still affect movement choices and create feedback effects. So, the argument does not need this condition to be true, and hence it isn't a necessary assumption.

Option D says that urban movement is uniform and predictable, leaving little room for interdependent behaviour. If this were true, cascading effects would be unlikely, but the passage clearly assumes the opposite: that behaviour is responsive and interlinked. Since the argument does not require uniformity and would actually be undermined by it, this assumption is also not necessary.

Hence, option B is the correct choice.

VIDEO SOLUTION

13. **3421**

The four sentences talk about how women are socialised to hold back anger, how this affects their relationship with aggression, how letting go of that restraint can feel liberating, and how dealing with anger shapes character and life for everyone, not just women. To figure out the correct order, we need to see how the ideas logically build.

Sentence **3** is the natural starting point because it introduces the central observation that women are often taught to hold back anger, while men's aggression is more accepted. Sentence **4** comes next, giving a real-life example from a coach's point of view to show how deeply this lesson is internalised. Sentence **2** follows by describing what happens when these rules[taboos] are broken, explaining the feeling of liberation that comes from expressing anger. Sentence **1** is the best ending because it moves beyond the gender-specific discussion and generalises, saying that how we handle anger shapes our character and our overall quality of life. Hence, the correct sequence is **3421**.

VIDEO SOLUTION

14. **C**

The passage describes how ageing, a weaker immune system, and chronic disease are connected. It introduces senescent or "zombie" cells, explains that younger people clear them more easily, and shows that this process slows with age. As a result, these cells build up, causing ongoing inflammation and damage to healthy tissue. Based on this, we consider each option individually.

Option A is too limited. It correctly says that senescent cells are malfunctioning and present in the body, but it misses the main point: as people age, these cells are not cleared as well, they build up, and this leads to inflammation.

Option B points out the differences between younger and older immune systems and mentions apoptosis, but it wrongly states that zombie cells 'resist' apoptosis. The passage actually focuses on the immune system becoming less effective, not the cells resisting. It also leaves out the important role of inflammation and chronic disease.

Option C covers the whole chain described in the passage. It links ageing to less effective apoptosis, explains how zombie cells build up, and includes their role in inflammation and chronic disease. This matches both the structure and the focus of the original text.

Option D is incorrect and somewhat out of the scope of the passage. The passage is about metabolically active senescent cells, not dead cells, and it does not say that inflammation directly weakens the immune system as this option suggests.

So, option C best sums up the main idea of the passage.

VIDEO SOLUTION

15. **A**

The last paragraph discusses the social and professional ties between nineteenth-century physicians and lawyers, such as their class, race, gender, institutional affiliations, and access to power. In this elite group, their interactions centred on a shared professional issue: how to define and enforce criminal responsibility, a topic on which they often disagreed.

Option A best explains this connection. Specifically, the paragraph states that, although these men had much in common, they were "divided by contests over the borders of criminal responsibility," both within and between their professions. This wording shows that debates over the limits of criminal responsibility were central to how these groups interacted in courtrooms and intellectual circles.

In contrast, the other options do not describe the connection between these men in this paragraph. Option B refers to an earlier time when family and neighbours judged sanity, before medical experts were involved in court. Option C mentions traits that alienists started to medicalise, but not what linked lawyers and doctors socially or professionally. Lastly, option D explains how jurors infer mental states in the first paragraph, not the elite networks or professional debates in the last paragraph.

VIDEO SOLUTION

16. B

The passage explicitly defines who alienists were in the nineteenth century: “physicians who specialized in the study of madness and the care of the insane... often called alienists or mad doctors.” They are described as the predecessors of modern psychiatrists, neurologists, and psychologists.

Option B matches this description. It correctly identifies both the profession (physicians) and their focus (madness and care of the insane), just as the passage describes.

The other options misread or exaggerate the term. Option A is too vague and does not mention the medical role described in the passage. Options C and D wrongly link “alienist” to immigrants or extraterrestrials, which the passage does not suggest. Therefore, option B is the correct answer.

[VIDEO SOLUTION](#)

17. D

The passage mainly discusses how mental state is judged in criminal law, how insanity is defined and assessed, and how these ideas affect legal responsibility and punishment. It often highlights the challenge of understanding mental states, the legal definition of responsibility as liability to punishment, and how medical and legal views on insanity became linked in the nineteenth century.

Option D best matches the ideas presented in the passage. *Judgement* refers to how jurors and officials make inferences. *Insanity* is the medical idea discussed and expanded by experts. *Punishment* links directly to Stephen’s idea of responsibility as “liability to punishment.” *Responsibility* is the key idea that connects law and medicine throughout the passage.

The other options include terms that are either not central to the discussion or do not encapsulate the ideas in the passage as well as D does. Option A mentions *business*, which only comes up as part of social ties, not as a main point. Option B lists *patronage* and *belief*, which are more about background networks than the main ideas of the passage. Option C combines ideas like empathy, which fits, with *patriotism*, which is only mentioned socially and not as part of the main analysis. So, option D is the set that is conceptually closest to the passage.

[VIDEO SOLUTION](#)

18. A

The sentence lists factors that made defendants seem *unlike* (different from) the officials judging them. This shows how fragile and subjective judgments about mental state could be. The word “confession” is included with disease, gender, and race, which are all seen as forms of perceived difference, not actions in a trial.

Option A best explains this use. In this context, “confession” means *religious confession or religious affiliation*. Like race or gender, it could set a defendant apart from officials in cultural or social ways. In the nineteenth century, religious identity often marked someone as different, and the sentence uses it in that sense.

The other options misunderstand the term. Option B wrongly treats “confession” as a verb meaning to claim an identity, which the sentence does not support. Option C does not make sense in the context, as it brings in the idea of a false confession, which is not mentioned. Option D takes “confession” to mean admitting a violent act, but this does not fit the sentence’s grammar/structure or the way the terms are grouped. So, option A is the correct answer.

[VIDEO SOLUTION](#)

19. 1

The sentences focus on one main idea: how people may have originally seen and experienced the Bayeux Tapestry, especially through *physical immersion* and *interaction* with art before the Renaissance. Most of the sentences work together to explain a scholarly argument about how the tapestry was arranged in space and how viewers engaged with it through their senses.

Sentences 3, 2, 5, and 4 are closely connected. Sentence 3 introduces Linda Neagley’s idea that people before the Renaissance interacted with art not just by looking, but also through physical movement and touch. Sentence 2 uses this idea to suggest, based on Gale Owen-Crocker, that the Bayeux Tapestry may have been hung in a square, with viewers standing in the middle. Sentence 5 adds a practical detail, saying the tapestry was likely hung at eye level to support this kind of viewing. Sentence 4 then explains that this setup would create an immersive space like in the 11th century, where scenes reflect each other and shape how viewers experience the story through their senses.

Sentence 1, on the other hand, is different from the rest. It makes a historical point, saying the tapestry was an “obvious way to tell people about the downfall of the English and the rise of the Normans.” This sentence, unlike the rest, does not connect logically with the discussion of display arrangement and sensory engagement developed in the other sentences. It introduces a different line of thought and therefore does not fit the coherent paragraph formed by the remaining four. Therefore, sentence 1 is the odd sentence out.

[VIDEO SOLUTION](#)

20. D

The passage presents an overview of different views on how income inequality relates to economic growth. The passage begins by explaining three ways inequality might help growth: investment indivisibilities, incentive effects, and concentrated corporate governance, which are backed by evidence, though there are signs this may change in the long run. The passage then moves on to discuss evidence from different countries and reasons why inequality could hurt growth over time [such as weaker human capital, etc.] Among the options, Option D best reflects both the possible benefits and the time-related caution in the passage.

The other options either exaggerate the claim or capture only a part of the argument. Option A focuses only on the difference between short- and long-term effects, which is just one part of the passage, not the main point. Option B is inaccurate because it says the passage only looks at certain channels and ignores opposing evidence, but the passage actually covers both sides. Option C is also misleading because it claims the author argues that inequality speeds up growth, when the passage really just presents different findings in a neutral way. The correct answer is therefore option D.

[VIDEO SOLUTION](#)

21. A

The passage states: “when democratization extends suffrage to a wider class of people, the increased political power of low- and middle-income voters results in broader support for income redistribution and social welfare expansion.” In the context of the passage, democratization refers specifically to the expansion of suffrage, i.e., extending voting rights to a broader segment of the population, thereby increasing the political influence of lower- and middle-income groups. It implies greater political inclusion and wider distribution of electoral power. Based on this, let’s consider each option individually.

Option A describes a move toward authoritarianism, where fewer people can vote, and opposition parties are removed. This is the direct opposite of expanding political participation, as it limits voting rights and competition. So, it matches the opposite of democratization.

Option B talks about corporate donations, public funding, and calls the system an oligarchy. While oligarchy means power is concentrated, limiting donations and using public funding are reforms that can exist in a democracy. So, this option does not clearly show the opposite of democratization as described in the passage.

Option C mentions participatory budgeting and recall elections, both of which increase civic participation. Even though the press release calls them “totalitarianism,” these actions actually show more democratization, not the opposite.

Option D talks about term limits and stronger judicial review, but says these lead to more autocratic rule. However, term limits and judicial review are usually ways to support democracy. The claim about autocracy is made, but the actions described do not clearly undermine democratization.

Therefore, option A is the correct answer here.

VIDEO SOLUTION

22. C

The three-part case explains how inequality may promote growth. The purpose of presenting these mechanisms is to specify the conditions under which inequality can support growth, particularly in the short run. This is best captured by option C, because it reflects those precise channels: high sunk costs, incentive alignment, and concentrated ownership, and does not overstate the claim.

The other options fail because Option A turns a conditional argument into a universal one; Option B distorts the financial-institution argument and adds a claim not made; and Option D directly contradicts the governance mechanism described in the passage.

VIDEO SOLUTION

23. A

The passage says that “economic performance is determined by the unobservable level of effort that agents make,” and if pay does not reflect performance, it “will fail to elicit optimum effort.” The passage concludes that some income inequality can help growth by motivating workers and encouraging innovation. The idea here is that linking pay differences to performance encourages people to work harder and be more productive. Option A fits this idea best because paying rewards for proven performance connects pay to results. This strengthens incentives and supports some inequality for efficiency reasons.

The other options do not work. Option B gives higher incomes without linking them to performance, so it does not solve the moral hazard problem. Option C bases wages on tenure rather than output, weakening effort incentives. Option D is about concentrated stock ownership, which is part of the governance argument, not the incentive argument. So, option A is the best answer.

VIDEO SOLUTION

24. A

The passage talks about a key issue in modern art: a globalised world lets artists find inspiration in many cultures, but this also brings the risk of cultural appropriation. The author explains that respectful and thoughtful inspiration can help people appreciate other cultures. On the other hand, cultural appropriation is not just borrowing, but borrowing without understanding, respect, or acknowledgement, which can lead to commodification, stereotypes, and bigger problems in society, like reinforcing power imbalances. Based on this, let’s consider each option individually.

Option A clearly shows both sides of the argument. It mentions the global context, separates respectful inspiration from appropriation that lacks acknowledgement, and highlights the passage’s focus on bigger social issues like power imbalances. This matches the passage’s main points well.

Option B points out the main difference between respectful inspiration and appropriation, but it is too general. It treats the issue as just a matter of “deciding” and leaves out the effects of appropriation and its impact on society, which are key parts of the passage.

Option C gets the author’s view wrong by suggesting that drawing from different cultures ‘always’ results in instances of both inspiration and cultural appropriation. The passage does not say this is inevitable. Instead, it says appropriation depends on how the borrowing happens. Also, “artists *must* draw from diverse cultural influences” is a bit strong, and it is also not implied in the passage.

Option D doesn’t present the complete argument, as it does not mention the global context or the idea of wider power imbalances, so it is a bit more limited in scope compared to the passage.

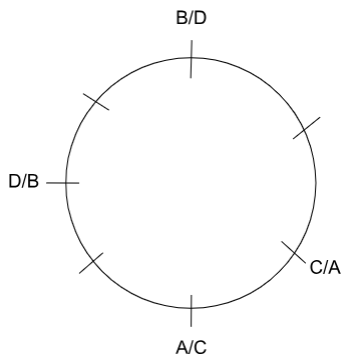
Overall, option A is the best choice because it covers the global context, the ethical difference between inspiration and appropriation, and the wider social effects the author talks about.

VIDEO SOLUTION

LRDI

Explanation [25 - 28]:

Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,

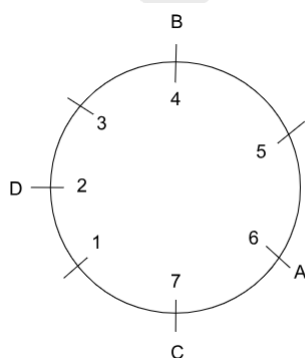


This has 4 possible combinations.

25.4

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



So, Bashir's initial position is chair 4.

Hence, the correct answer is 4.

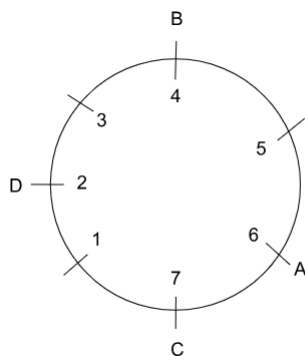
[VIDEO SOLUTION](#)



26. D

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

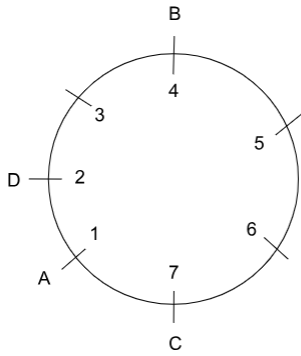
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

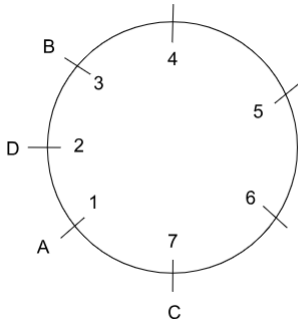
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



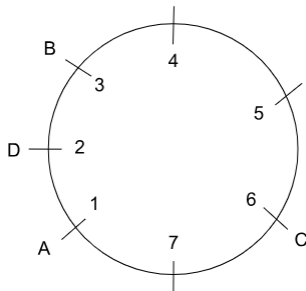
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



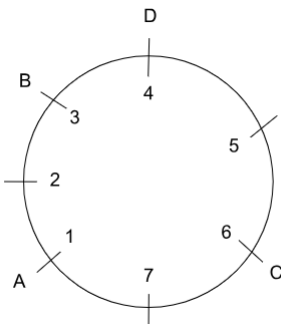
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

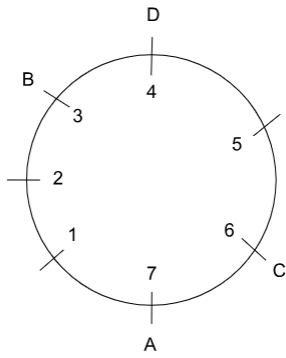
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

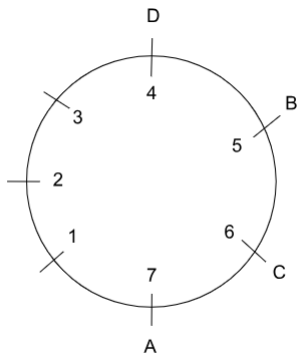
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



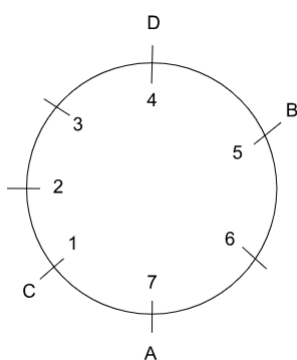
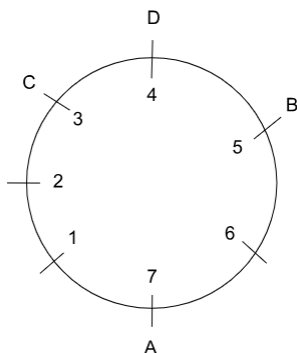
Turn 6: Bashir moves

Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.



After turn 3, chair 4 is occupied by no one.

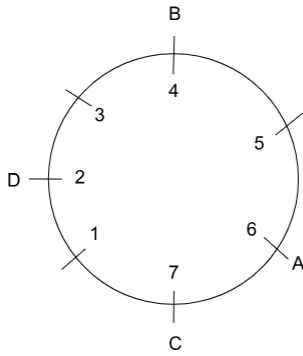
Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

27. A

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

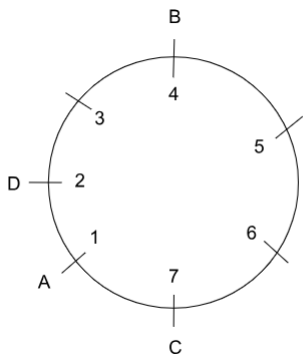
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

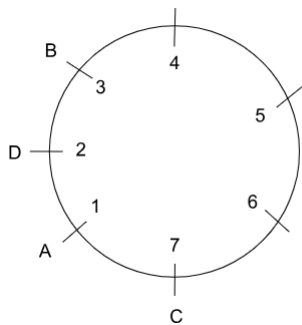
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



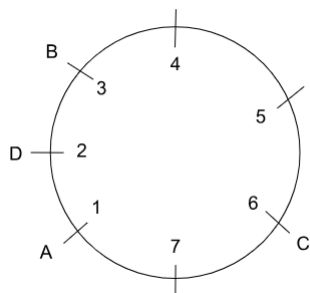
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



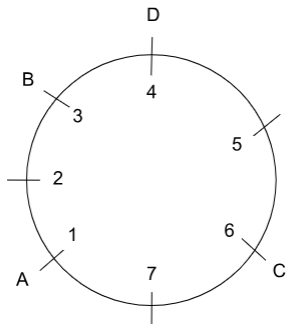
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

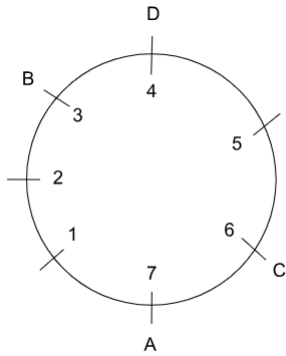
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

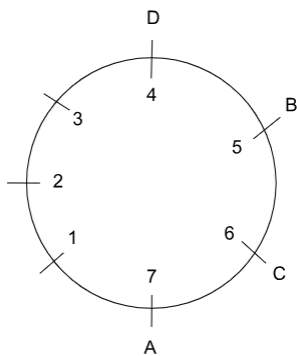
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



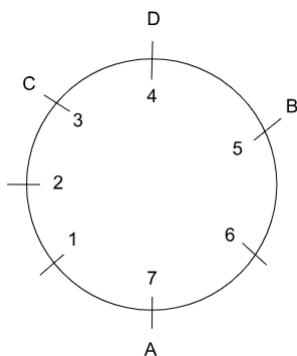
Turn 6: Bashir moves

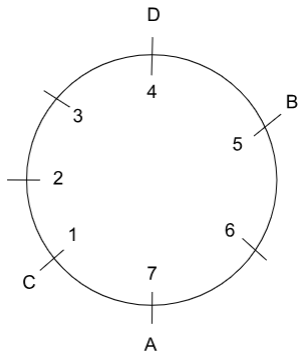
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





The chairs occupied after turn 6 are 4, 5, 6 and 7.

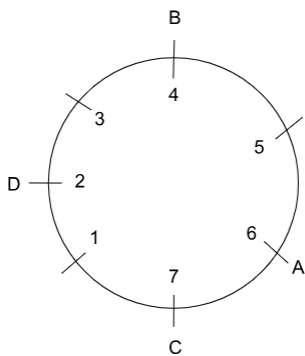
Hence, the correct answer is option A.

[VIDEO SOLUTION](#)

28. **B**

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

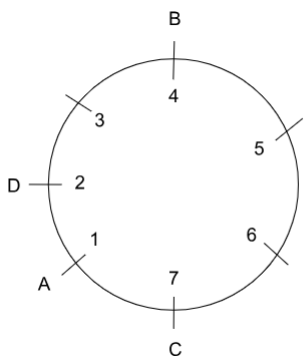
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

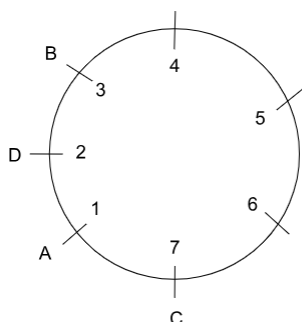
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



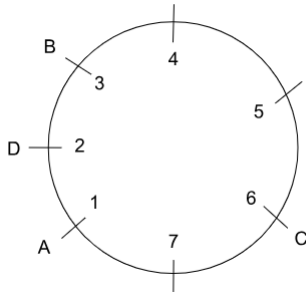
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



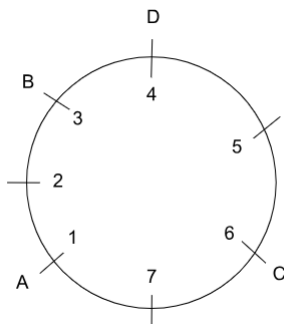
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

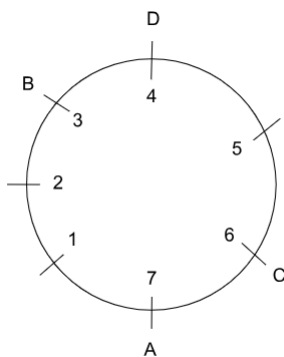
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

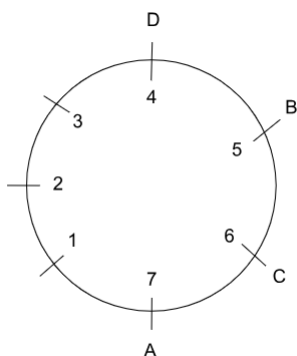
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



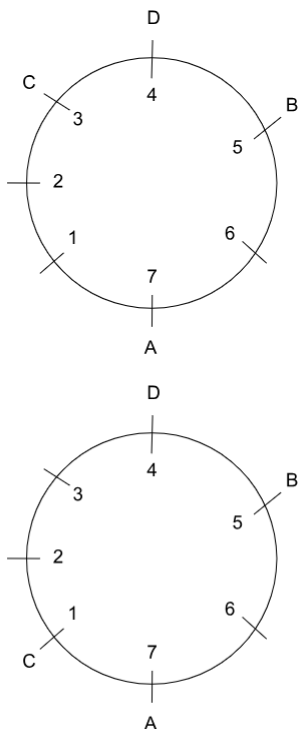
Turn 6: Bashir moves

Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.



In either of the cases after turn 7, the seat adjacent to Bashir is occupied only by Davies, and the other one is empty.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

29.4

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Ekalavya's score after quarter 2 is 4.

Hence, the correct answer is 4.

[VIDEO SOLUTION](#)

30.0

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already

know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

As we can see in the table, no employee has changed their group more than once.

Hence, the correct answer is 0.

[VIDEO SOLUTION](#)



CAT Daily Targets

Step by step, heading towards 99+ percentile



31.5

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novie after Q1, and B cannot go back to Elite after Q2 and come back to Novie after Q3 because B came to Novie in Q1, and A came to Novie in Q3, so C has to come to Novie after Q3 to be present in Novie at the start of Q4 as per the table. Therefore, we can conclude that C came to Novie after Q3, and B stayed in Novie in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novie after Q2. So, we can conclude that D stayed in Novie during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novie after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novie in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novie after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novie from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novie after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novie and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novie after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novie, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Bunty's score after quarter 3 is 5.

Hence, the correct answer is 5.

VIDEO SOLUTION

32.4

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novie after Q1, and B cannot go back to Elite after Q2 and come back to Novie after Q3 because B came to Novie in Q1, and A came to Novie in Q3, so C has to come to Novie after Q3 to be present in Novie at the start of Q4 as per the table. Therefore, we can conclude that C came to Novie after Q3, and B stayed in Novie in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novie after Q2. So, we can conclude that D stayed in Novie during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novie after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novie in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novie after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novie from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novie after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novie and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novie after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novie, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

The cumulative score of 4 employees can be determined with certainty at the end of quarter 3.

Hence, the correct answer is 4.

 VIDEO SOLUTION

33. D

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novie after Q1, and B cannot go back to Elite after Q2 and come back to Novie after Q3 because B came to Novie in Q1, and A came to Novie in Q3, so C has to come to Novie after Q3 to be present in Novie at the start of Q4 as per the table. Therefore, we can conclude that C came to Novie after Q3, and B stayed in Novie in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novie after Q2. So, we can conclude that D stayed in Novie during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novie after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novie in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novie after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novie from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

I. Asha received a rating of 2 in Quarter 1 - This is not necessarily true, as it can also be 3 in Quarter 1.

II. Asha received a rating of 1 in Quarter 2. - This is definitely true, as Asha received a rating of 1 in Quarter 1.

Only II is definitely correct.

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

Explanation [34 - 37]:

The values given in both charts together are represented in the table below, with the import tariff percentages charged by each of the five countries on the others represented as a percentage, and the import tariffs levied by each country on other countries are represented in brackets(in billion USD).

	US	France	India	Japan	UK
US		20% (6)	40% (4.5)	30% (7)	30% (3)
France	30% (5.5)		40% (6.5)	30% (4)	40% (4.5)
India	40% (9)	50% (5)		50% (3.5)	20% (5)
Japan	40% (4)	30% (3)	50% (8)		40% (3)
UK	20% (2.5)	30% (4)	30% (3)	40% (6)	

34. C

Japan's export to India would be the same as India's import from Japan.

India is charging a 50% tariff on Japan, and the tariff by India on Japan equals 3.5 billion USD.

So, 50% of the imports equals 3.5 billion USD.

$$\frac{50}{100} \times \text{Imports} = 3.5$$

$$\text{Imports} = 3.5 \times 2 = 7 \text{ billion USD.}$$

The value of imports by India from Japan = Japan's export to India = 7 billion USD.

Hence, the correct answer is option C.

[VIDEO SOLUTION](#)

35. B

Option A) Exports by Japan to the UK would be the same as the UK's imports from Japan.

The UK is charging a 40% tariff on Japan, and the tariff imposed by the UK on Japan equals 6 billion USD.

So, 40% of the imports equals 6 billion USD.

$$\frac{40}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 3 \times 5 = 15 \text{ billion USD.}$$

The value of imports by the UK from Japan = Japan's export to the UK = 15 billion USD.

Option B) Imports by the US from France.

The US is charging a 20% tariff on France, and the tariff by the US on France equals 6 billion USD.

So, 20% of the imports equals 6 billion USD.

$$\frac{20}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 6 \times 5 = 30 \text{ billion USD.}$$

The value of imports by the US from France = 30 billion USD.

Option C) Exports by France to Japan would be the same as Japan's imports from France.

Japan is charging a 30% tariff on France, and the tariff by Japan on France equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by Japan from France = France's export to Japan = 10 billion USD.

Option D) Imports by France from India.

France is charging a 40% tariff on India, and the tariff by France on India equals 6.5 billion USD.

So, 40% of the imports equals 6.5 billion USD.

$$\frac{40}{100} \times \text{Imports} = 6.5$$

$$\text{Imports} = 6.5 \times 2.5 = 16.25 \text{ billion USD.}$$

The value of imports by France from India = 16.25 billion USD.

Out of all the options, the value of imports by the US from France is the highest.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)



36. **B**

Trade surplus/trade deficit of India with the UK = Exports from India to the UK - Imports from India to the UK

Exports by India to the UK would be the same as the UK's imports from India.

The UK is charging a 30% tariff on India, and the tariff imposed by the UK on India equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by the UK from India = India's export to the UK = 10 billion USD.

Imports by India from the UK can be calculated as,

India is charging a 20% tariff on the UK, and the tariff imposed by India on the UK equals 5 billion USD.

So, 20% of the imports equals 5 billion USD.

$$\frac{20}{100} \times \text{Imports} = 5$$

$$\text{Imports} = 5 \times 5 = 25 \text{ billion USD.}$$

The value of imports by India from the UK = 25 billion USD.

We can clearly see that the Imports are greater than the exports for India from the UK. So, the trade deficit can be calculated as,

Trade deficit = Exports from India to the UK - Imports from India to the UK = 10 billion USD - 25 billion USD = -15 billion USD.

So, there is a deficit of 15 billion USD.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

37. **D**

For France - Trade surplus/trade deficit of France with the US = Exports from France to the US - Imports from the US to France

Exports by France to the US would be the same as the US's imports from France.

The US is charging a 20% tariff on India, and the tariff imposed by the US on France equals 6 billion USD.

So, 20% of the imports equals 6 billion USD.

$$\frac{20}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 6 \times 5 = 30 \text{ billion USD.}$$

The value of imports by the US from France = France's export to the US = 30 billion USD.

Imports by France from the US can be calculated as,

France is charging a 30% tariff on the US, and the tariff imposed by France on the US equals 5.5 billion USD.

So, 30% of the imports equals 5.5 billion USD.

$$\frac{30}{100} \times \text{Imports} = 5.5$$

$$\text{Imports} = 5.5 \times 3.34 = 18.34 \text{ billion USD.}$$

The value of imports by France from the US = 18.34 billion USD.

We can clearly see that the Imports are less than the exports for France from the US. So, the trade surplus can be calculated as,

Trade surplus = Exports from France to the US - Imports from France to the US = 30 billion USD - 18.34 billion USD = 11.67 billion USD.

So, there is a surplus of 11.67 billion USD.

For the UK - The Trade surplus/trade deficit of the UK with the US = Exports from the UK to the US - Imports from the US to the UK

Exports by the UK to the US would be the same as the US's imports from the UK.

The US is charging a 30% tariff on the UK, and the tariff imposed by the US on the UK equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by the US from the UK = The UK's export to the US = 10 billion USD.

Imports by the UK from the US can be calculated as,

The UK is charging a 20% tariff on the US, and the tariff imposed by the UK on the US equals 2.5 billion USD.

So, 20% of the imports equals 2.5 billion USD.

$$\frac{20}{100} \times \text{Imports} = 2.5$$

$$\text{Imports} = 2.5 \times 5 = 12.5 \text{ billion USD.}$$

The value of imports by the UK from the US = 12.5 billion USD.

We can clearly see that the Imports are greater than the exports for the UK from the US. So, the trade deficit can be calculated as,

Trade deficit = Exports from the UK to the US - Imports from the UK to the US = 10 billion USD - 12.5 billion USD = -2.5 billion USD.

So, there is a deficit of 2.5 billion USD.

We can see that France has a surplus and the UK has a deficit with the UK.

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

38. B

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	$c = a$
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ --(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

Occupancy factor of segment D - E can be calculated as,

$$\text{Occupancy factor} = \frac{140}{200} \times 100 = 70\%$$

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

39.50

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a, b, c, x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ ---(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ ---(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ ---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked from station A to E from the above table is 50.

Hence, the correct answer is 50.

[VIDEO SOLUTION](#)

40.80

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \quad \text{---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

Number of tickets booked from station C = $20 + 60 = 80$.

Hence, the correct answer is 80.

[VIDEO SOLUTION](#)



41.40

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a, b, c, x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ ---(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ ---(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are a + 30.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ ---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of a > 30.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when a + 30 is a multiple of 4, and the only value out of the above values that satisfies the condition is when a = 50, as at a = 40 and 60, the value of a + 30 is not a multiple of 4.

We can conclude that the value of a = 50, and substituting in (1), we get the value of b = 30.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked to station C = $50 + 40 = 90$

The number of tickets booked to station D = $30 + 20 = 50$

Difference = $90 - 50 = 40$.

Hence, the correct answer is 40.

[VIDEO SOLUTION](#)

42. 60

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ ---(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ ---(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ ---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked for exactly one segment can be calculated using values from the above table as,

$$\text{Tickets booked for exactly one segment} = A - B + B - C + C - D + D - E = 0 + 40 + 20 + 0 = 60.$$

Hence, the correct answer is 60.

[VIDEO SOLUTION](#)

Explanation [43 - 46]:

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

$$6 + \text{Badal} + \text{Clive} + 11 + 9 = 40$$

$$\text{Badal} + \text{Clive} = 14$$

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

43.7

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be $1 + 1 + 2 + 2 = 6$. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as $2 + 2 + 1 + 3 = 8$. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

Hence, the correct answer is 7.

[VIDEO SOLUTION](#)

44. D

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be $1 + 1 + 2 + 2 = 6$. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

Alia and Badal are the only people with an equal number of taps.

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

45. A

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be $1 + 1 + 2 + 2 = 6$. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as $2 + 2 + 1 + 3 = 8$. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C		1/3				8
D	3	2	3		3	11
E		3/1				9
TAPS RECEIVED	9	8	7	8	8	40

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be $9 - 3 - 1 = 5$. So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's

question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is $8 - 1 = 7$. So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2/1	2	1/2	6
C	2	1/3		2/3	2/1	8
D	3	2	3		3	11
E	3	3/1	1/2	3/2		9
TAPS RECEIVED	9	8	7	8	8	40

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2	2	1	6
C	2	1		3	2	8
D	3	2	3		3	11
E	3	3	1	2		9
TAPS RECEIVED	9	8	7	8	8	40

Clive's answer to Ehsan's question is No, which is denoted by 2.

Hence, the correct answer is option A.

[VIDEO SOLUTION](#)



46.6

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be $1 + 1 + 2 + 2 = 6$. It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C						8
D	3	2	3		3	11
E						9
TAPS RECEIVED	9	a		a	a	40

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as $2 + 2 + 1 + 3 = 8$. We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B						6
C		1/3				8
D	3	2	3		3	11
E		3/1				9
TAPS RECEIVED	9	8	7	8	8	40

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be $9 - 3 - 1 = 5$. So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is $8 - 1 = 7$. So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2/1	2	1/2	6
C	2	1/3		2/3	2/1	8
D	3	2	3		3	11
E	3	3/1	1/2	3/2		9
TAPS RECEIVED	9	8	7	8	8	40

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,

	Q1(A)	Q2(B)	Q3(C)	Q4(D)	Q5(E)	TAPS
A		2	1	1	2	6
B	1		2	2	1	6
C	2	1		3	2	8
D	3	2	3		3	11
E	3	3	1	2		9
TAPS RECEIVED	9	8	7	8	8	40

The total number of Yes responses is equal to the number of 1's in the table, which is 6.

Hence, the correct answer is 6.

Quant

47. B

First function $f(x) = x^2 - 4cx + 8$

For this function $a > 0$, so minimum value will occur at $x = -\frac{b}{2a} = -\left(-\frac{4c}{2}\right) = 2c$

So, the minimum value of the function is $= 2c^2 - 4c(2c) + 8c = -4c^2 + 8c$

Second function $g(x) = -x^2 + 3cx - 2c$

For this function $a < 0$, so maximum value will occur at $x = -\frac{b}{2a} = -\frac{(-3c)}{2} = \frac{3c}{2}$

So, the maximum value of the function is $= -\left(\frac{3c}{2}\right)^2 + 3c\left(\frac{3c}{2}\right) - 2c = \frac{9c^2}{4} - 2c$

So, as per the given condition,

$$\frac{9c^2}{4} - 2c < -4c^2 + 8c$$

$$\text{or, } \frac{9c^2}{4} + 4c^2 < 8c + 2c$$

$$\text{or, } \frac{25c^2}{4} < 10c$$

$$\text{or, } \frac{5c^2}{4} < 2c$$

$$\text{or, } 5c^2 < 8c$$

$$\text{or, } 5c^2 - 8c < 0$$

$$\text{or, } c\left(c - \frac{8}{5}\right) < 0$$

$$\text{or, } 0 < c < \frac{8}{5}$$

So, the value of c which lies in this range is $\frac{1}{2}$

VIDEO SOLUTION

48. D

According to question, Shruti travels a distance of 224 km in four parts for a total travel time of 3 hours.

Given, in the first part time required is 30 minutes.

Also, the time taken to cover these four parts follow arithmetic progression.

So, let us say the times be 30 minutes, $(30+d)$ minutes, $(30+2d)$ minutes and $(30+3d)$ minutes

$$\text{So, } 30 + (30 + d) + (30 + 2d) + (30 + 3d) = 180$$

$$\text{or, } 6d = 180 - 120$$

$$\text{or, } 6d = 60$$

$$\text{or, } d = \frac{60}{6} = 10 \text{ minutes.}$$

So, the time required in these four parts are 30 minutes, 40 minutes, 50 minutes and 60 minutes respectively.

Now, in the first part, speed is 960 metres per minute.

Speed in the four parts is also in arithmetic progression.

Let's say the speeds be $(960 + x)$, $(960 + 2x)$, $(960 + 3x)$ metres per minute

So, total distance covered $= 960 \times 30 + (960 + x) \times 40 + (960 + 2x) \times 50 + (960 + 3x) \times 60$ metres

$$\text{So, } 960 \times 30 + (960 + x) \times 40 + (960 + 2x) \times 50 + (960 + 3x) \times 60 = 224000$$

$$\text{or, } 960(30 + 40 + 50 + 60) + 40x + (2x)(50) + (3x)(60) = 224000$$

$$\text{or, } 172800 + 320x = 224000$$

$$\text{or, } 320x = 224000 - 172800 = 51200$$

$$\text{or, } x = \frac{51200}{320} = 160$$

So, the speed in the fourth part is $960 + 3x = 960 + 3 \times 160 = 1440$ metres per minute

Time in the fourth part is 60 minutes

So, distance covered in the fourth part $= 1440 \times 60 = 86400$ metres

So, option D is the correct answer.

VIDEO SOLUTION

49. 6

According to question, in the given 3-digit number N, the digits are non-zero and distinct.

So, the possible digits in 3-digit number = 2, 3, 5, 6, 7, 8

It is also given that only one of the digits is a prime number.

So, the minimum possible value of $N = 268$

(2 is the smallest prime digit, and the non-prime digits has to be 6 and 8)

$$\text{Now, } 268 = 4 \times 67 = 2^2 \times 67$$

$$\text{So, the number of factors} = (2 + 1)(1 + 1) = 3 \times 2 = 6$$

[VIDEO SOLUTION](#)

50. A

For $n=3,4,5$ and $x \in [n, n+1)$ we have $\lfloor x \rfloor = n$, so the equation

$$\lfloor x^2 \rfloor = \lfloor x \rfloor^2 = n^2$$

$$x^2 \in [n^2, n^2 + 1), \text{ i.e. } x \in [n, \sqrt{n^2 + 1})$$

Thus for $3 \leq x \leq 6$

$$S = [3, \sqrt{10}) \cup [4, \sqrt{17}) \cup [5, \sqrt{26}) \cup 6$$

Option B and C have $\sqrt{10}$ included, which is not part of the original set. And Option D has $\sqrt{18}$. So, it is not possible.

Option A is the answer.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51. 15

Let the number of stocks B hold be x

So, number of stocks C hold is $20 - x$

$$\text{So, } 10 \times 120 + 90 \times x + 150 \times (20 - x) = 3300$$

$$\text{or, } 1200 + 90x + 150(20 - x) = 3300$$

$$\text{or, } 1200 + 90x + 3000 - 150x = 3300$$

$$\text{or, } 4200 - 60x = 3300$$

$$\text{or, } 60x = 900$$

$$\text{or, } x = \frac{900}{60} = 15$$

So, the number of shares of stock that B hold is 15.

[VIDEO SOLUTION](#)

52. A

Given expression is $\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} < \{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$,

$$\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} = \{3^1 \times 3^4 \times 3^9 \dots \times 3^{400}\}$$

Sum of square of n natural numbers is $\frac{n \cdot (n+1) \cdot (2n+1)}{6}$

$$= 3 \frac{(20 \cdot 21 \cdot 41)}{6} = 3^{2870}$$

On right hand side of inequality we have $\{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$

$$= 3^{21} \times 3^{22} \times \dots \times 3^{20+m} = 3^{21+22+\dots+20+m}$$

Using the sum of the first (n) natural numbers,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$21 + 22 + \dots + (20 + m)$$

$$= 1 + 2 + \dots + (20 + m) - (1 + 2 + \dots + 20)$$

$$1 + 2 + \dots + (20 + m) = \frac{(20+m)(21+m)}{2}$$

$$1 + 2 + \dots + 20 = \frac{20 \cdot 21}{2} = 210$$

$$\text{So, } 21 + 22 + \dots + (20 + m)$$

$$= \frac{(20+m)(21+m)}{2} - 210$$

$$\text{Expanding, } (20 + m)(21 + m) = m^2 + 41m + 420$$

$$\text{Thus, } \frac{m^2 + 41m + 420}{2} - 210$$

$$= \frac{m^2 + 41m}{2}$$

Since the bases are equal, we must compare the powers.

$$2870 < \frac{m^2 + 41m}{2} \Rightarrow 5740 < m(m + 41)$$

Here, we can put in the option to check the minimum value that satisfies the inequality.

56: We get $5740 < 5264$. This is false

57: We get $5740 < 5586$. This is false

58: We get $5740 < 5742$. This is the minimum possible value.

[VIDEO SOLUTION](#)

53. D

For base of log in range $1/4 \in (0, 1)$ and $\log_{1/4}(x) > 0$ is true only if $0 < x < 1$.

For integer n , $x = n^2 - 7n + 11$ is an integer, so it cannot lie strictly between 0 and 1.

So, there is no integer value for which this inequality is satisfied.

[VIDEO SOLUTION](#)

54. 16

It is given, $x > y \geq 3$ and $x + y < 14$

Now for $y = 3$, the values of x can be 4, 5, 6, 7, 8, 9, 10 (7 cases)

Then for $y = 4$, the values of x can be 5, 6, 7, 8, 9 (5 cases)

Then for $y = 5$, the values of x can be 6, 7, 8 (3 cases)

Then for $y = 6$, the values of x will be 7 (1 case)

So, total number of cases = $1+3+5+7=16$ cases

[VIDEO SOLUTION](#)

55. A

Let the principal be ₹ P and rate of interest be $r\%$.

$$\text{Now, } 13920 = P + \frac{P \times r \times 3}{100}$$

$$\text{or, } 13920 - P = \frac{P \times r \times 3}{100} \text{ ---->(1)}$$

$$\text{Also, } 18960 = P + \frac{P \times r \times 13}{100 \times 2}$$

$$18960 - P = \frac{P \times r \times 13}{100 \times 2} \text{ ---->(2)}$$

Dividing eqn(1) by eqn(2),

$$\frac{13920 - P}{18960 - P} = \frac{3}{\frac{13}{2}} = \frac{6}{13}$$

$$\text{or, } (13920 - P) 13 = (18960 - P) 6$$

$$\text{or, } 13920 \times 13 - 13P = 18960 \times 6 - 6P$$

$$\text{or, } 13920 \times 13 - 18960 \times 6 = 13P - 6P$$

$$\text{or, } 180960 - 113760 = 7P$$

$$\text{or, } 67200 = 7P$$

$$\text{or, } P = \frac{67200}{7} = 9600$$

Putting this in equation (1),

$$13920 - 9600 = \frac{9600 \times r \times 3}{100}$$

$$\text{or, } 4320 = 96 \times 3r$$

$$\text{or, } r = \frac{4320}{96 \times 3} = 15$$

So, rate percent is 15%

Now if the same sum had been invested for 2 years at the same rate as before but with interest compounded every 6 months, amount =

$$9600 \left(1 + \frac{15}{2 \times 100}\right)^4 = 9600 \left(1 + \frac{7.5}{100}\right)^4$$

$$\text{So, interest} = 9600 \left(1 + \frac{7.5}{100}\right)^4 - 9600$$

$$= \text{Rs } 3220.50$$

$$= \text{Rs } 3221$$

So, the total interest earned is Rs 3221.

[VIDEO SOLUTION](#)

56. D

200 L with (30%) acid $\Rightarrow$ acid = $0.30 \times 200 = 60L$

Replace 20% with water: remaining acid = $60 \times (1 - 0.20) = 60 \times 0.8 = 48L$.

Replace 10% with pure acid: After removing 10% of the mixture, the acid becomes $48 \times 0.9 = 43.2L$, then adding back 20 L pure acid $\Rightarrow$ acid = $43.2 + 20 = 63.2L$.

Replace 15 with water: acid left = $63.2 \times (1 - 0.15) = 63.2 \times 0.85 = 53.72L$.

Final concentration = $\frac{53.72}{200} = 0.2686 \approx 26.86$.

The answer is 27%

VIDEO SOLUTION

57. 55

Let say number of girls be g and number of boys be b .

If 40% of the girls left, remaining number of girls = $0.6g$

Also if 60% of the boys left, remaining number of boys = $0.4b$

or, $0.6g = 0.4b + 8$

or, $6g = 4b + 80$

or, $3g = 2b + 40$

So, the possible values of (b,g) are: (13,22),(16,24),(19,26),(22,28),(25,30),....

Now, $0.6g$ and $0.4b$ has to be an integer.

So, for this g and b has to be a multiple of 5

So, $b = 25$ and $g = 30$

So, minimum possible number of students = $25 + 30 = 55$

VIDEO SOLUTION

58. A

Number of ways to choose a sandwich = 5C_1 ways

Number of ways to choose a bread = 4C_1 ways

Number of ways to choose bread size = 2C_1 ways

Number of ways to choose sauces = ${}^6C_0 + {}^6C_1 + {}^6C_2 = 1 + 6 + 15 = 22$ ways

So, number of different ways = ${}^5C_1 \times {}^4C_1 \times {}^2C_1 \times 22 = 5 \times 4 \times 2 \times 22 = 880$ ways.

VIDEO SOLUTION

59. A

The sum of all the elements in the given set = Sum of first 29 odd numbers = $29^2 = 841$

Let's assume that k is the m_{th} term. Sum of terms less than k = sum of first $(m-1)$ odd numbers = $(m-1)^2$

$$841 - m^2 = (m-1)^2$$

$$841 - m^2 = m^2 - 2m + 1$$

$$840 - 2m^2 + 2m = 0$$

$$m^2 - m - 420 = 0$$

$$(m-21)(m+20) = 0$$

$$m = 21 - 20$$

$$m = 20. \text{ And the 20th term is } 2 \times m + 1 = 41$$

VIDEO SOLUTION

60. A

Let's assume that Total work = 1

If the work is entirely carried by each one of them it would cost

$$\text{Arun: } 2160 \times 24 = 51840$$

$$\text{Varun: } 2400 \times 21 = 50400$$

$$\text{Tarun: } 2160 \times 15 = 32400$$

The cheapest worker is Tarun. He can finish the work in 15 days. But we need the work completed in 10 days.

Tarun does $10 \times \frac{1}{15} = \frac{2}{3}$

$1 - \frac{2}{3} = \frac{1}{3}$

Complete the remaining (1/3) work using the next cheapest worker, which is Varun.

Days needed: $\frac{x}{21} = \frac{1}{3} \Rightarrow x = 7$

Total cost = Amount paid for Tarun + Amount paid for Varun = $10 \times 2160 + 7 \times 2400 = 38400$

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61. 900

Let the amounts invested in Stocks be S, Bonds B, and Gold be G

Given that $S + B + G = 100000, B = 0.25G$

$0.10S + 0.06B + 0.08G = 8200$

Substitute $S = 100000 - B - G = 100000 - 1.25G$

$0.10(100000 - 1.25G) + 0.06(0.25G) + 0.08G = 8200$

$10000 - 0.125G + 0.015G + 0.08G = 8200$

$10000 - 0.03G = 8200$

$0.03G = 1800 \Rightarrow G = 60000$

$B = 0.25G = 15000$

Gain from bonds = $0.06 \times 15000 = 900$

[VIDEO SOLUTION](#)

62. C

Given, $a - 6b + 6c = 4 \rightarrow (1)$

and, $6a + 3b - 3c = 50 \rightarrow (2)$

Multiplying eqn (1) with $x, ax - 6bx + 6cx = 4x$

And multiplying eqn (2) with $y, 6ay + 3by - 3cy = 50y$

So, $x + 6y = 2$ and $3y - 6x = 3$

So, $x + 6y = 2 \rightarrow (3)$ and $-2x + y = 1 \rightarrow (4)$

Multiplying eqn (4) with 6 and subtracting from eqn (3),

$13x = -4$

So, $x = -\frac{4}{13}$

Putting the value of x in equation (4),

$y = \frac{5}{13}$

So, the final answer is $4x + 5y$

$= 4\left(-\frac{4}{13}\right) + 50\left(\frac{5}{13}\right) = -\frac{16}{13} + \frac{250}{13} = \frac{234}{13} = 18$

So, correct answer is 18.

[VIDEO SOLUTION](#)

63. D

Given $(P(-3,-2)), (Q(1,-5)),$ and $(R(9,1)).$

For parallelogram (PQRS), $S = P + R - Q = (-3,-2) + (9,1) - (1,-5) = (5,4)$

Diagonal SQ passes through $S(5,4)$ and $Q(1,-5).$

Slope $m = \frac{-5 - 4}{1 - 5} = \frac{9}{4}$

Equation of SQ is $y - 4 = \frac{9}{4}(x - 5)$

At the x-axis, $y = 0: -4 = \frac{9}{4}(a - 5)$

$-16 = 9(a - 5)$

$9a = 29$

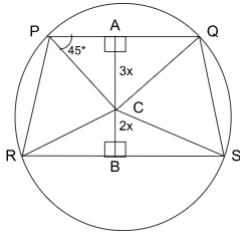
$a = \frac{29}{9}$

$$\text{So, } a = \frac{29}{9}$$

VIDEO SOLUTION

64. C

The radius of the circle is given to be $6\sqrt{2}$ cm. We are also given that the ratio of CA to CB is 3:2. So, if we assume the value of CA to be $3x$, then the value of CB becomes $2x$. We are also given that $\angle PQC = 45^\circ$ and since PCQ is an isosceles triangle, $\angle CPQ = 45^\circ$ and $\angle PCQ = 90^\circ$.



So, the triangle PCQ is right-angled at C. The value of PQ can be calculated using Pythagoras' theorem as,

$$PQ^2 = PC^2 + QC^2$$

$$PQ^2 = (6\sqrt{2})^2 + (6\sqrt{2})^2 = 72 + 72 = 144$$

$$PQ = 12 \text{ cm}$$

$$AQ = \frac{PQ}{2} = \frac{12}{2} = 6 \text{ cm}$$

In triangle ACQ,

$$AC^2 + AQ^2 = CQ^2$$

$$(3x)^2 + (6)^2 = (6\sqrt{2})^2$$

$$9x^2 + 36 = 72$$

$$9x^2 = 36$$

$$x^2 = 4$$

$$x = 2$$

$$CB = 2x = 2 \times 2 = 4 \text{ cm}$$

In triangle BCS,

$$BC^2 + BS^2 = CS^2$$

$$(4)^2 + (BS)^2 = (6\sqrt{2})^2$$

$$16 + (BS)^2 = 72$$

$$BS^2 = 56$$

$$BS = 2\sqrt{14} \text{ cm}$$

$$RS = 2 \times BS = 2 \times 2\sqrt{14} = 4\sqrt{14} \text{ cm}$$

The area of the quadrilateral PQRS can be calculated as,

$$\text{Area} = \frac{1}{2} \times AB \times (PQ + RS) = \frac{1}{2} \times 10 (12 + 4\sqrt{14}) = 20 (3 + \sqrt{14}) \text{ cm}^2$$

Hence, the correct answer is option C.

VIDEO SOLUTION

65. D

Let $M=13k$, $A=9k$.

$$\text{After 21 students were moved } \frac{13k-21}{9k+21} = \frac{19}{14} \Rightarrow 14(13k-21) = 19(9k+21)$$

$$\text{So } 182k - 294 = 171k + 399 \Rightarrow 11k = 693 \Rightarrow k = 63$$

So, the number of students in the morning and afternoon shifts are $819 - 21 = 798$, $567 + 21 = 588$ respectively.

Let's assume that $3t$ and $8t$ students joined the respective sessions.

$$\frac{798+3t}{588+8t} = \frac{5}{4} \Rightarrow 4(798+3t) = 5(588+8t)$$

$$\text{So } 3192 + 12t = 2940 + 40t \Rightarrow 28t = 252 \Rightarrow t = 9$$

$$\text{Number of new students} = 11t = 99$$

VIDEO SOLUTION

CAT Expected Questions PDF



Let the diagonals of the rhombus be d_1 and d_2

We know that the Area $= \frac{1}{2}d_1d_2 = 396 \implies d_1d_2 = 792$

For a rhombus with side $a=36$, the diagonals intersect at right angle. Giving a right-angle triangle with the side as hypotenuse.

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2 \implies \frac{d_1^2 + d_2^2}{4} = 36^2 = 1296$$

$$d_1^2 + d_2^2 = 5184$$

We want $|d_1 - d_2|$

$$(d_1 - d_2)^2 = d_1^2 + d_2^2 - 2d_1d_2 = 5184 - 2 \cdot 792 = 5184 - 1584 = 3600$$

$$|d_1 - d_2| = \sqrt{3600} = 60$$

[VIDEO SOLUTION](#)

67.3

The given quadratic equation is $x^2 - 5x + k = 0$

Now, discriminant $D = 5^2 - 4k = 25 - 4k$

Now, it is given the equation must have integer roots.

So, $25 - 4k$ has to be a perfect square.

We need to find non-negative integer values of k

Now, for $k = 0$, $D = 25 - 4 \times 0 = 25$, is a perfect square

For $k = 4$, $D = 25 - 4 \times 4 = 25 - 16 = 9$, is a perfect square

For $k = 6$, $D = 25 - 4 \times 6 = 1$, is a perfect square

So, there are three non negative integer values of k .

So, correct answer is 3.

[VIDEO SOLUTION](#)

68.175

Cost price of each chair = 100

For 13 chairs, total cost = $13 \times 100 = 1300$

Profit = 26%, so total revenue

$$1.26 \times 1300 = 1638$$

We were told that this amount is equal to the discounted price of 12 chairs. So the discounted selling price per chair = $SP_{\text{disc}} = \frac{1638}{12} = 136.5$

Discount offered = 22%, so:

$$MP = \frac{136.5}{0.78} = 175$$

[VIDEO SOLUTION](#)